PIPELINE AUDIT AND HEALTH REPORT

Automotive: body and engine marriage line

Painted bodies and tested engines arrive on two supply lines and meet in an assembler that needs one of each. Final inspection (a custom quality_gate) sends defects to a rework bay and scraps cars that fail twice. The paint booth is slowest and breaks down, so it limits the whole plant. Units count bodies and engines separately, so each car is 2 units.

Generated 2026-10-13 19:48 | simulated 8.0 h (warm-up 60 min) | seed 81 | 10 replications

HEALTH SCORE

78 /100

Grade C

Acceptable: workable, but there are clear risks to address before scaling up.

Critical 0 | High 0 | Medium 5 | Low 2 | Info 1

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
54.6 units/h	438 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
13.1 units	860 s	paint

Executive summary

Primary bottleneck: 'paint' (71% busy, 21% down, index 92). Sources are held back up to 30% of the time. Run what-if to measure which lever recovers the most throughput.

What to fix first

- Medium (supply): The line cannot absorb the offered supply, so everything upstream of the constraint spends time waiting.
- Medium (paint): Plan capacity before demand grows further. Measured best lever: 10% faster processing at 'paint' (+7.1% throughput).
- Medium (engine_test): Speed up or add capacity downstream of this station, or enlarge the buffer after it.
- Medium (paint): Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.

Best measured improvement

Best lever: 10% faster processing at 'paint' -> +4.2 units/hour (7.1%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Medium	Flow	**supply**	Upstream supply is held back up to 30% of the time by a full line. Action: The line cannot absorb the offered supply, so everything upstream of the constraint spends time waiting.	-5
Medium	Capacity	**paint**	The bottleneck is 92% utilised. Limited slack for demand growth or disturbances. Action: Plan capacity before demand grows further. Measured best lever: 10% faster processing at 'paint' (+7.1% throughput).	-4
Medium	Flow	**engine_test**	Blocked 44% of the time: finished parts cannot leave because the next buffer is full. Action: Speed up or add capacity downstream of this station, or enlarge the buffer after it.	-3
Medium	Reliability	**paint**	Down 21% of the time (6 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Medium	Reliability	**paint**	Every unit passes through this station, and it can fail. A single failure stops the whole plant. Action: Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.	-4
Low	Confidence	**throughput**	Throughput 95% CI is +-5.2% of the mean. Action: Use a longer run if you need tighter estimates.	-1
Low	Model	**run length**	The measured window (7.0 h) is only 29x the cycle time. Action: Aim for at least 30x the cycle time so averages settle.	-2
Info	Capacity	**paint**	Constraint of the line: 92% utilised. Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'paint' (+7.1% throughput).	

Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
bodies	source	-	-	-	-	-
paint bottleneck	machine		71.2	0.0	20.8	2.9 / 4
drying	delay		71.9	2.1	0.0	0.9 / 3
engines	source	-	-	-	-	-
engine_test	machine		46.5	43.9	0.0	2.7 / 4

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
marriage	assembler		53.4	0.0	0.0	3.9 / 6
final_qc	quality_gate		36.9	0.0	0.0	0.8 / 4
rework_bay	machine		49.6	0.0	0.0	1.4 / 6
dealer	sink	-	-	-	-	-
scrap_yard	sink	-	-	-	-	-

Work in process over time

Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	62.51 units/h	+- 3.25	4.55	5.2%
Average WIP	13.79 units	+- 0.40	0.56	2.9%
Cycle time	792.60 s	+- 40.40	56.49	5.1%
Arrivals lost	0.00 %	+- 0.00	0.00	0.0%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
paint	60 min	8.0 min	88.2%	20.8%	6

Single points of failure

All units must pass through: paint, engine_test, marriage, final_qc. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 449 = completed 438 + scrapped 2 + lost 0 + in system 9
Little's Law	PASS	L = 13.12, lambda x W = 13.10 (gap 0.1%)
Run length	WARN	window = 29 x cycle time (30x or more recommended)
Warm-up	PASS	first 60 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 58.9 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
paint	10% faster processing	63.0	**+4.2**	+7.1%
paint	+2 buffer slots	61.4	**+2.6**	+4.4%
paint	20% shorter repairs	61.2	**+2.4**	+4.0%
marriage	+2 buffer slots	59.0	**+0.1**	+0.2%
marriage	10% faster processing	58.9	**+0.0**	+0.0%
rework_bay	+2 buffer slots	58.9	**+0.0**	+0.0%
rework_bay	10% faster processing	58.8	**-0.1**	-0.2%

Best lever: 10% faster processing at 'paint' -> +4.2 units/hour (7.1%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
bodies	source	interarrival=exponential(90), mix={"body":1}, on_blocked=wait
paint	machine	capacity=3, mtbf=3600, mttr=480, process=normal(95,8)
drying	delay	capacity=6, transit=120
engines	source	interarrival=exponential(90), mix={"engine":1}, on_blocked=wait
engine_test	machine	capacity=3, process=normal(60,6)
marriage	assembler	capacity=3, kit={"body":1,"engine":1}, output_type=car, process=normal(70,8)
final_qc	quality_gate	capacity=4, max_reworks=2, pass_to=dealer, process=normal(40,8), rework_to=rework_bay, scrap_to=scrap_yard, threshold=0.10

NODE	TYPE	EFFECTIVE PARAMETERS
rework_bay	machine	capacity=3, process=normal(300,60)
dealer	sink	(defaults)
scrap_yard	sink	counts_as=scrapped

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
Medium	**supply**	Upstream supply is held back up to 30% of the time by a full line.	-5
Medium	**paint**	The bottleneck is 92% utilised. Limited slack for demand growth or disturbances.	-4
Medium	**engine_test**	Blocked 44% of the time: finished parts cannot leave because the next buffer is full.	-3
Medium	**paint**	Down 21% of the time (6 failures).	-3
Medium	**paint**	Every unit passes through this station, and it can fail. A single failure stops the whole plant.	-4
Low	**throughput**	Throughput 95% CI is +-5.2% of the mean.	-1
Low	**run length**	The measured window (7.0 h) is only 29x the cycle time.	-2
	Total deductions		**-22**
	Health score		**78**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.